

Shashank Karki

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Education

Virginia Polytechnic Institute and State University PhD. in Geospatial and Environmental Analysis Thesis: Enhancing Urban spaces with GeoAI and visual language models Relevant Coursework: Urban Computing, Big Data Engineering, GenAI for Social Sciences	Blacksburg, VA 2024-Present
Virginia Polytechnic Institute and State University Meng. Computer Science and Applications Relevant Coursework: Machine Learning, Social Media Analytics, Data Structures and Algorithms	Blacksburg, VA 2025-Present
Virginia Polytechnic Institute and State University Masters in Geography, GPA: 4.0 Thesis: Tracking Human Movement and Occupancy Indoors Using Terrestrial Lidar Advisors: Dr. Thomas Pingel and Dr. Timothy Baird Relevant Coursework: Programming for GIS, Enterprise GIS, Analysis of Spatial Data, Advanced Analysis in GIS, Advanced Topics in Remote Sensing	Blacksburg, VA 2022-2024
Kathmandu University Bachelor of Engineering in Geomatics Engineering, GPA: 3.84 Thesis: Analyzing the Effect of Image Overlap, Number of GCPs & Distribution Pattern on the Overall Accuracy of UAV Photogrammetry Relevant Coursework: Python Programming for GIS, Photogrammetry, Spatial Database Management System, Remote Sensing, Geographic Information System	Dhulikhel, Nepal 2014-2018

Experience

Virginia Polytechnic Institute and State University Graduate Teaching Assistant, Department of Geography - Delivered instructional support for various GIS courses, including Principles of GIS, Modeling with GIS, and Programming for GIS, as well as Remote Sensing. - Provided lab assistance and support for the lead instructor. - Enhanced skills in teaching, verbal communication, critical reasoning, and leadership.	Blacksburg, VA Jan 2022 - Dec 2024
Virginia Polytechnic Institute and State University Graduate Research Assistant, Department of Geography <i>Assessment of Accessibility and Transportation Network Modeling</i> - Conducted comprehensive accessibility assessments for dental care services across Virginia. - Managed and analyzed public transit data using General Transit Feed Specification (GTFS) files. - Developed transportation network models utilizing Rapid Realistic Routing (R5R) in R. - Employed Pandas and GeoPandas for extensive travel time data analysis and management. <i>Lidar Sensor Data Analysis and Object Detection</i> - Managed and calibrated lidar sensors for optimal data collection. - Analyzed lidar data to detect space use patterns and identify moving objects. - Developed deep learning models for object detection in lidar feeds. - Created and implemented accuracy metrics, including bidimensional regression, chamfer distance, and chamfer ratio, for method comparison. <i>Green Exposure Index Calculation and Image Segmentation</i> - Calculated the Green Exposure Index for home to work commutes in Virginia.	Blacksburg, VA Jun 2023 - Aug 2023 Jun 2023 - Aug 2023 May 2023 - Jun 2023

- Utilized Google Directions API and Street View API for route calculation and imagery.
- Applied PixelLib for image segmentation and greenery ratio calculations.
- Performed data aggregation and analysis using Pandas

Virginia Polytechnic Institute and State University

Blacksburg, VA

Graduate Research Assistant, Department of Fish and Wildlife Conservation

May 2022 - Aug 2022

- Computed Land Surface Temperature data at a 30m resolution.
- Assessed habitat changes in Bog Turtle populations from the 1980s-2020s.
- Utilized Google Earth Engine for satellite imagery analysis and management.
- Managed and performed spatial overlay operations on over 10 TBs of satellite imagery using High-performance Computing (HPC) resources.

Kathmandu University

Dhulikhel, Nepal

Graduate Teaching Assistant, Department of Geomatics Engineering

Mar 2019 – Oct 2021

- Conducted lectures and practical classes on GIS, Theory of Errors & Adjustments, and Control Survey.
- Led survey camps, instructing and managing a group of 60 students.
- Maintained and managed equipment in both the Survey and Photogrammetry Labs.

GeoNode Nepal, funded by the World Bank Group

Remote

Spatial Database Manager

Nov 2019 – Nov 2021

- Managed and updated a free spatial data repository for Nepal, available at GeoNode Nepal.
- Developed expertise in geospatial data management over geoportals through hands-on experience and research.

Nepal Electricity Authority

Kathmandu, Nepal

UAV and DGPS Surveyor

Jun 2021 – Jul 2021

- Conducted UAV surveys using Mavic 2 Pro for a prospective hydropower plant site in Dolpa, Nepal.
- Processed aerial imagery to generate Ortho mosaics, Digital Terrain Models (DTM), and Digital Surface Models (DSM) using Pix4D.
- Differential GPS Surveys carried out to establish accurate ground control points for geospatial data correction.

Energy Development Council Nepal

Kathmandu, Nepal

Graduate Teaching Assistant, Department of Geomatics

Jun 2018 – Aug 2028

- Estimated theoretical hydropower potential within specified regions.
- Performed hydrological modeling to assess water flow and resource availability.
- Utilized ArcHydro Tools in ArcMap for spatial analysis.
- Integrated precipitation datasets from WorldClim for comprehensive environmental modeling.

Research

Publications

- (In preparation) Wang, C., **Karki, S.**, & Zhang, M. (2026). Shifts in dental service areas following Medicaid policy change: A statewide analysis of Virginia Medicaid claims. (Expected submission: April 2026).
- (In preparation) Kim, J., **Karki, S.**, Gyanwali, S., Chandramouli, V., David-John, B., & Bloom, M. (2026). From presence to perception: Quantifying human engagement with urban greenery using an AI-powered wearable eyetracking device. Expected submission: March 2026.
- (In preparation) **Karki, S.**, Kim, J., Hankey, S., Ye, X., & Oliver, R. (2026). Generative AI for urban and spatial planning: A systematic review of methods, data practices, and spatial reasoning. Expected submission: March 2026.

- (Under review) **Karki, S.**, Kim, J., Oliver, R., & Jang, K. M. (2025). An AI-driven framework for enhancing urban walkability using image analysis, augmentation, and community-centric improvements. Manuscript in review.
- Sherman, Z., **Karki, S.**, Brickhouse, T., Vujicic, M., Nasseh, K., Wang, C., Kim, H., Kim, J., & Zhang, M. (2026). Geospatial access to oral health care in Virginia: Insights from driving and public transit analysis. *The Journal of the American Dental Association*, 157(1), 36-46.e6.
<https://doi.org/10.1016/j.adaj.2025.09.003>
- Casado, G. L., **Karki, S.**, John, V., Kim, J., & Zhang, M. (2025). Transit-Based Model to Evaluate Geospatial Accessibility of Eye Care in Virginia. *Investigative Ophthalmology & Visual Science*, 66(8), 2100–2100.
- Flack, A. H., Pingel, T. J., Baird, T. D., **Karki, S.**, & Abaid, N. (2025). Lidar-Based Detection and Analysis of Serendipitous Collisions in Shared Indoor Spaces. *Remote Sensing*, 17(18), 3236.
<https://doi.org/10.3390/rs17183236>
- Zhang, T., Yu, H., & **Karki, S.** (2025). Empowering entrepreneurs: age, telework, and geographic context in transitioning to knowledge-based self-employment. *Entrepreneurship & Regional Development*, 1-37.
- Kim, J., **Karki, S.**, Brickhouse, T., Vujicic, M., Nasseh, K., Wang, C., & Zhang, M. (2025). Navigating Disparities in Dental Health—A Transit-Based Investigation of Access to Dental Care in Virginia. *Community Dentistry and Oral Epidemiology*, 53(1), 117-124.
- **Karki, S.**, Pingel, T. J., Baird, T. D., Flack, A., & Ogle, T. (2024). Enhancing Digital Twins with Human Movement Data: A Comparative Study of Lidar-Based Tracking Methods. *Remote Sensing*, 16(18), 3453.
- Gyanwali, S., **Karki, S.**, Jang, K. M., Crawford, T., Zhang, M., & Kim, J. (2024). Implications for spatial nonstationarity and the neighborhood effect averaging problem (NEAP) in green inequality research: evidence from three states in the USA. *Journal of Geographical Systems*, 1-24.
- Bista, N., Mahat, D., Manandhar, S., Regmi, B., Panday, U. S., & **Karki, S.** (2021). Analyzing Trend and Pattern of Agricultural Drought: A Case Study of Karnali and Sudurpashchim Provinces. *Journal on Geoinformatics, Nepal*, 20(1), 1–8. <https://doi.org/10.3126/njg.v20i1.39470>
- Shrestha, A., Ghimire, S., Pokharel, A., Dhimi, S., **Karki, S.**, Awasthi, B. and Acharya, T., (2021). Landslide susceptibility mapping in Budhi Gandaki River subbasin using frequency ratio and statistical Information index. *Bulletin of Nepal Hydrogeological Association*, 6.
- Awasthi B., **Karki S.**, Regmi P., Dhimi D.S., Thapa S., Panday U.S. (2020) Analyzing the Effect of Distribution Pattern and Number of GCPs on Overall Accuracy of UAV Photogrammetric Results. In: Jain K., Khoshelham K., Zhu X., Tiwari A. (eds) *Proceedings of UASG 2019*. UASG 2019. Lecture Notes in Civil Engineering, vol 51. Springer, Cham. https://doi.org/10.1007/978-3-030-37393-1_29
- Shrestha, R., Zevenbergen, J., Panday, U. S., Awasthi, B., and **Karki, S.**, (2019). Revisiting the Current UAV Regulations in Nepal: A Step towards Legal Dimension for UAV's Efficient Application, *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, XLII-5/W3, 107–114, <https://doi.org/10.5194/isprs-archives-XLII-5-W3-107-2019>
- Awasthi, B., **Karki, S.**, Regmi, P., Dhimi, D. S., Thapa, S., & Panday, U. S. (2019). Assessing The Overall Accuracy of UAV Photogrammetry: A Methodology for Different Combination of Imagery Parameters and Terrain Types. *Journal of Land Management and Geomatics Education*, 01(1), ISSN: 2676-1408, 22–32.

Presentations

- **Karki, S.**, Oliver, R., Hankey, S., Ye, X., & Kim, J., (2026). Generative AI in Spatial and Built Environment Planning: A Systematic Review of Methods, Data Practices, and Spatial Reasoning. American Association of Geographers (AAG) Annual Meeting, San Francisco, CA., March 17–23.
- **Karki, S.**, Kim, J., Oliver, R., & Jang, K. M. (2025). GeoAI twins for walkable cities: Modeling urban perception with generative image analysis. Presented at the 2025 CGA Conference: The Geography of Digital Twins, Cambridge, Massachusetts, May 22–24.
- **Karki, S.**, Kim, J., & Oliver, R. (2025). An AI-driven framework for enhancing urban walkability using image analysis, augmentation, and community-centric improvements. American Association of Geographers (AAG) Annual Meeting, Detroit, MI., March 24–28.
- **Karki, S.**, Pingel, T.J., Flack, A., & Baird, T.D. (2023). Aggregating Multi-Sensor Terrestrial Lidar Data for Analyzing Human Behavior. Presented at the 2023 ASPRS International Technical Symposium, June 12-16.

Invited Talks

- **Karki, S.** (2023). Capturing and Modelling Movement and Behavior in a Digital Twin. SPARC Workshop on Digital Twins in Geospatial Research, Arizona State University, February 27–28.
- **Karki, S.** (2025). An AI-driven framework for enhancing urban walkability using image analysis, augmentation, and community-centric improvements. GEOG 4304/5304: Geospatial Analysis of Mobility, Virginia Tech. November

Posters

- **Karki, S.**, Casado, G. L., John, V. J., Kim, J., & Zhang, M. (2025, May). Eyes on transit: Disparities in accessing eye care by public transportation in Virginia [Poster presentation]. Carilion Research Day, Virginia Tech Carilion School of Medicine, Roanoke, VA.
- **Karki, S.**, Jang, K. M., Oliver, R., & Kim, J. (2025, April). Enhancing urban walkability with AI: Image processing, augmentation, and community-driven solutions. OGIS Symposium 2025, Office of GIS and Remote Sensing, Virginia Tech
- **Karki, S.**, Flack, A., Pingel, T.J., and Baird, T.D. (2023). Navigating the Indoor Frontier: Uncovering Movement and Occupancy Patterns with Terrestrial Lidar. Virginia Tech OGIS Symposium. April 21.
- **Karki, S.**, Wilson, B., Granger, A., Flack, A., Neal, C., Dressel, L., Carruthers, C., and Pingel, T. (2022). Indoor Mapping. Virginia Tech OGIS Symposium. April 9.
- **Karki, S.**, and Rijal, S. (2022). Linear Spectral Unmixing for Vegetation Estimates at Different Spatial Resolutions. Virginia Tech OGIS Symposium. April 9.
- **Karki, S.**, (2019). Non-Governmental GeoNode: A geospatial content management system for Nepal. Presented at the International Workshop on "Capacity Building and Education Outreach in Advanced Geospatial Technologies and Land Management", LMTC, Dhulikhel, Kavre, December 10-11.

Leadership & Activities

FOSS4G Asia 2021

Kathmandu, Nepal

Local Organizing Committee

Feb 2021

- Led the planning and execution of a conference schedule and venue management managing over 150 attendees.
- Coordinated with different departments, secured sponsorships, and managed a team of volunteers.
- Led a team of about 20 volunteers

International GIS Day 2017, Kathmandu University

Dhulikhel, Nepal

Event Coordinator

Mar 2017

- Led the development and preparation of the event program, ensuring a comprehensive and engaging schedule for attendees.
- Acted as the official host for the event, facilitating smooth transitions between segments and maintaining audience engagement

Honors & Awards

Induction into the Honor Society Of Phi Kappa Phi

Blacksburg, VA

Virginia Polytechnic Institute & State University

Apr 2023

- Recognition: Outstanding academic excellence
- Selection: By invitation, requiring nomination and chapter approval
- Eligibility: Top 10% of seniors and 7.5% of juniors, along with eligible graduate students, faculty, professional staff, and distinguished alumni

Best Paper Award

Roorkee, India

India Institute of Technology

Apr 2019

- Event: International Conference on Unmanned Aerial System in Geomatics 2019
- Date: April 06-07, 2019
- Location: IIT Roorkee
- Recognition: Outstanding contribution and presentation of paper

Academic Excellence Scholarship

Roorkee, India

Kathmandu University

2014-2018

- Institution: Department of Geomatics Engineering, Kathmandu University
- Semesters: 1, 3, 5, 6 and 8
- Recognition: Awarded for achieving the highest academic performance in the respective semesters during undergraduate studies

Skills & Interests [Note: Optional]

Technical:

- Geographic Information Systems (GIS):
 - Proficient: ArcGIS Suite (ArcMap, ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise), QGIS, ERDAS IMAGINE, Google Earth, Google Earth Engine
 - Familiar: Leaflet, GeoDa, ArcGIS Server, AutoCAD
- Photogrammetry:
 - Pix4D, WebODM, Agisoft Metashape
- Programming Languages:
 - Intermediate/Working Proficiency: Python, R
 - Beginner: C, C++, Java, JavaScript
 - Python Libraries: Numpy, Pandas, GeoPandas, Scipy
 - Development Environments: Spyder, Jupyter Notebook, JupyterLab, Google Colab
- Database Management:
 - Sqlite, PostGIS, MySQL, PostgreSQL, Apache Spark
- Web Mapping and Geospatial Web Development:
 - Mapbox, OpenLayers, Geoserver, PostGIS Web Services

- Version Control:
 - Git (GitHub)
- Cloud Platforms:
 - Amazon Web Services

Language: English (Fluent); Nepali (Native); Hindi (Intermediate)

Interests: Soccer, Gaming, Movies, Road trips